Roll No. | | | | | | | | ***Set-1***

END SEMESTER EXAMINATION 2025

Name of the Course: B.Tech Petroleum Engineering *Semester: V*

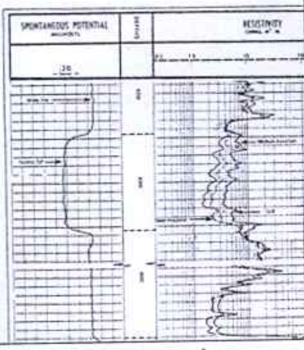
Name of the Paper: Well logging & Formation Evaluation *Course Code:* TPE-502

Time: Three Hours MM: 100

Note:

i. All questions are compulsory.
ii. Answer any two sub questions in each main question.
iii. Total marks for each main question is **twenty**
iv. Each Question carries **10 marks**

Q1		COs
a.	What is the difference between well logging & well log? What are the standard logging tools and their principle of measurements?	CO-1 CO-2
b.	Explain term vertical resolution and depth of investigation of standard logging tools used in well logging.	CO-1 CO-2
c.	What is the difference between well logging & mud logging? Which one is more accurate and reliable? Explain with reason.	CO-1 CO-2
Q2		
a.	Explain the theory of measurement of Gama Ray (GR) tool with diagram. How to identify sand, shale sequences based on GR log w.r.t. depth.	CO-3 CO-4
b.	Explain Caliper log and tool principle. Write the formula for calculation of mud cake thickness from Caliper log.	CO-2 CO-3
c.	Write formula for volume of shale estimation using GR Log. Solve for Vshale if GR Log= 90, GRShale= 200 & GR Clean= 50. Write the uses of GR log in sedimentary basin.	CO-2 CO-3
Q3		

a.	Explain the theory of measurement of SP log with tool diagram. From the figure draw Sand and shale base lines by drawing the same figure. Write the value of SP log at the points indicated by arrows. 	CO-3 CO-4
b.	Explain resistance & resistivity? Drive formula for equivalent resistance in series & parallel combinations with figure. Why is resistivity preferred for the measurement by logging tools?	CO-3
c.	Draw a neat diagram showing various zones created during drilling process? How are the resistivity tools designed to measure the resistivity in these zones? What are their limitations?	CO-3
Q4		
a.	How density tool measure formation density? Explain the principle of Density tool with figure.	CO-5
b.	Explain the atomic structure of an atom. How Neutron tool measure formation Porosity? What are the limitation of Neutron log?	CO-5
c.	Write the formula for Estimation of porosity from density log. In a sandstone formation, Matrix density = 2.65, Density log = 2.35, and Density Fluid = 1.0 Solve for Porosity =?	CO-4 CO-5
Q5		
a.	Illustrate the propagation of compressional and Shear waves with diagram. Write the operating principle of Sonic logging tools. What type of porosity this tool will measure?	CO-5
b.	Drive Wyllie time-average equation for prediction of porosity. Solve this equation for porosity if □t is 80 □s/ft and the matrix □tma □□55.5 us/ft and □tfluid□□189 us/ft. Write the limitation of sonic tool.	CO-5
c.	What are the different tools for identification of lithology? Explain with figures.	CO-6